

02 – Full System Loop

WORLD → ECR → Discovery → Causal certification → Machine Factory
... → Archive → Lineage → Certified matter → RCF → FPGA → Body → WORLD

Digital Chemistry

Coupled state-reaction rules on typed channels (IN_A, IN_B, OUT, H*).

Commit: sum contributions; state $\leftarrow \tanh(\text{state} + \text{delta})$.

Machine Factory

Search/mutate on substrate; promote only after causal/load-bearing tests.

Archive

Sole certified construction material; parents and certificates attached.

CLM / PCM

Compose and mount certified primitives; persistent body over archive matter.

ECR

Unknown-world act/intervene/certify; can promote into MF archive (v0.3 loop).

Faithful IR / CHC

Software semantics for hardware projection; gated RTL; bitstream emission.

RCF

Fixed host fabric; slots; certified packages only; hot-load/unload/gate; body bus.

What is frozen

Software discovery campaigns and Hop-2; direction is physical RCF + embodiment.